

5(a)

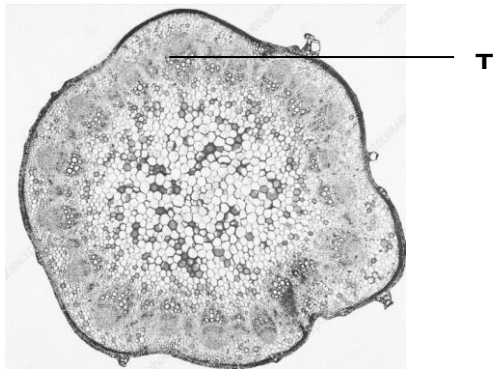
one mark for named pathways ;
one mark for outline of apoplastic pathway ;

2

name of pathway	outline of pathway
symplast / symplastic	movement of water from cell to cell via plasmodesmata
apoplast / apoplastic	movement of water via, cell walls / intercellular spaces

5(b)

label line labelled **T** to any area of xylem tissue ;
e.g.

1

Question	Answer	Marks
5(c)(i)	lysosome ;	1
5(c)(i)i	any three from: increased permeability to water ; increased, aquaporins / protein channels (in the tonoplast) ; water potential in vacuole is lower than that of cytoplasm ; A more ions in vacuole lower water potential more water enters (vacuole) by osmosis ; vacuole increases, in volume / pressure (until it bursts) ; AVP ; e.g. fewer transport proteins for exit of ions	3
5(c)(iii)	<i>allow two suggestions or a suggestion with further detail, for two marks</i> active site becomes complementary to substrate ; (because) bonds altered between R groups of amino acids ; A hydrogen / ionic, bonds removal of a portion that blocks the active site ; AW substrate can enter active site ; <i>idea of cofactor</i> / AW, associates with the enzyme ; enzymes becomes functional ; sets off an enzyme cascade / ref. cell signalling ; AVP ;	2

Question	Answer	Marks
5(d)(i)	any two from: caused by a pathogen ; <i>idea that</i> it is transmissible / passed from infected plant to uninfected plant ; results in harm being caused / AW ;	2
5(d)(ii)	protocist ; <u>Plasmodium falciparum</u> / <u>Plasmodium malariae</u> / <u>Plasmodium ovale</u> / <u>Plasmodium vivax</u> ; A <i>Plasmodium knowlesi</i>	2

Question	Answer	Marks
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